Lecture 8: Information

Zhixian Yu

University of Nottingham Ningbo China



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Lecture 8 Outline

- Incomplete information about quality or price
  - Lemon market
  - The Tourist-trap model
  - The Tourist-and-native model
- Asymmetric information between two players
- Bayesian persuasion
- Extra Reading:
  - Carlton and Perloff (1999) Modern industrial organization. Chapter 13
  - Akerlof (1970). The Market for "Lemons": Quality Uncertainty and the Market Mechanism. The Quarterly Journal of Economics.
  - Stiglitz (1979). Equilibrium in Product Markets with Imperfect Information. The American Economic Review.



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Introduction

- Consumers are imperfectly informed. They do not know the characteristics of all the products in the market or the prices at which they are available from all sellers.
- If consumers have incomplete information about **quality**, the market may not exist, or may only provide low-quality products.
- If consumers have incomplete information about **prices**, the market price will be higher, and there might be price dispersion.



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Incomplete Information About Quality

- Consumers often do not know how quality varies across brands in markets for the services of professionals (such as doctors, lawyers, plumbers, electricians, and economists), processed foods, used goods, and complex mechanical or electronic products. There is asymmetric information: one party (the seller) to a transaction knows a material fact (the quality of the good) that the other party (the buyer) does not. Asymmetric information about quality can result in either of two undesirable outcomes: an equilibrium may not exist, or if the equilibrium exists, resources are used less efficiently than they would be if there were perfect, symmetric information.

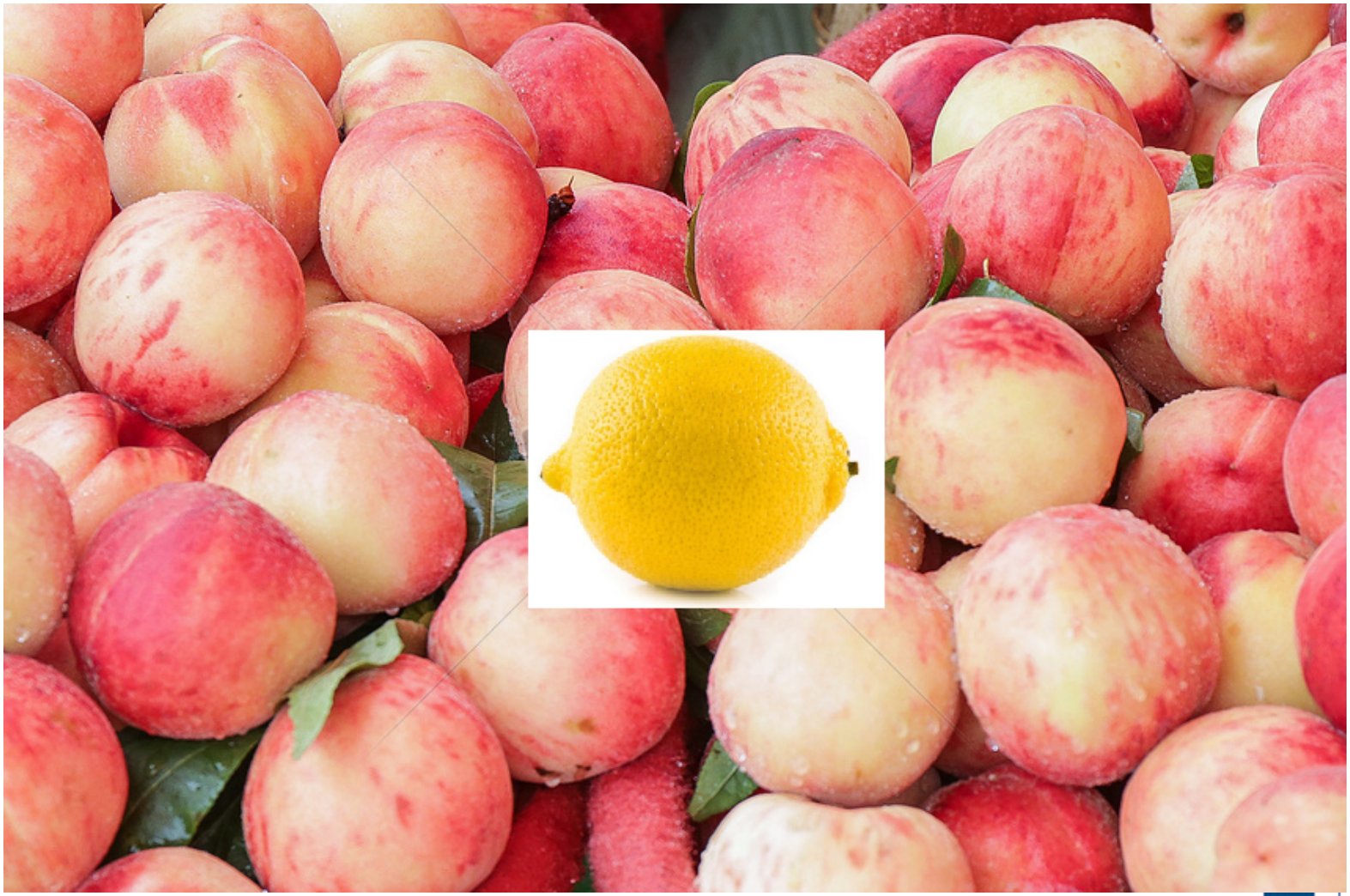
Market for Lemons

- The market for "lemons" (Akerlof 1978) is one of the simplest yet most important concepts in economics.
- "Lemons" represent low-quality products that sellers know about but buyers do not. Suppose the market has both lemon and non-lemon products, but buyers cannot tell the difference. In this case, the lemon sellers have incentives to pretend to be non-lemon sellers.
- When buyers realize that the market has lemons, they will refuse to pay the non-lemon price, and their price will be a mixture of lemon and non-lemon prices.
- This drives out the non-lemon sellers and further reduces buyers' expected price.
- This repetition process causes adverse selection to drive out all non-lemon products, leading to a market collapse.



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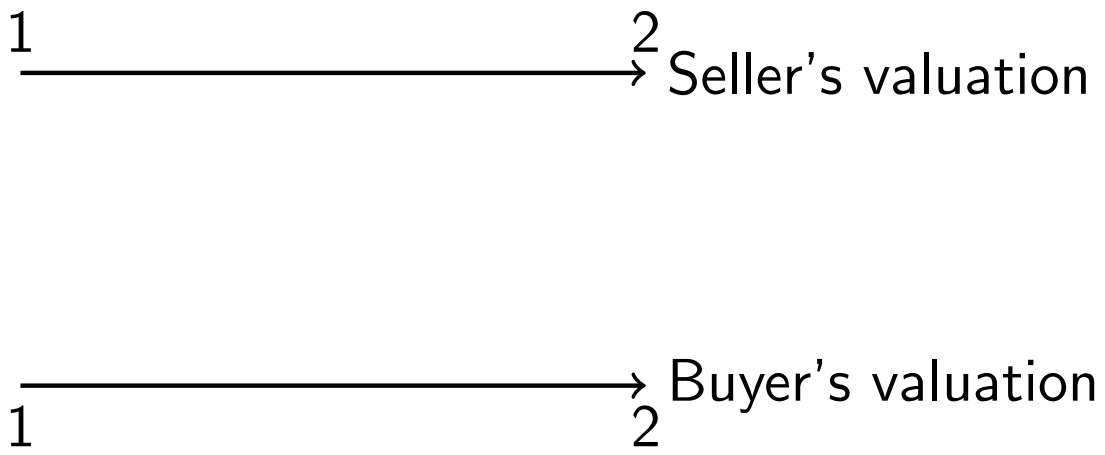
Can you find it?



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Market for Lemons

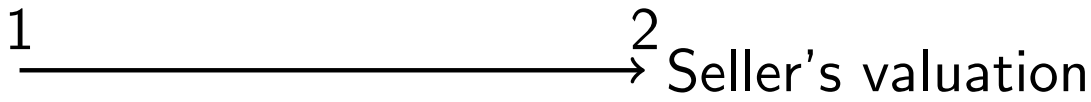
- The basic assumption is that the products have UNIFORMLY DISTRIBUTED qualities, and consumers pay based on their EXPECTED prices.
- Suppose that both the buyer and seller value lemons at 1 and peaches (the opposite of lemons) at 2. In this case, the market will have only lemons left.

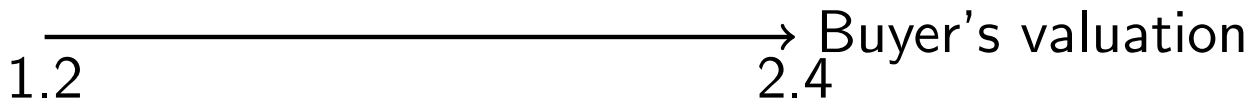


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Market for Lemons

- Of course, the situation is not always so hopeless. In markets, buyers usually have a higher valuation of the products than sellers, say, 20%. Therefore, when moving from lemons to peaches, the seller has a valuation of 1 to 2, while the buyer has a valuation of 1.2 to 2.4.
- What will be the final equilibrium?





- Now, can you think of some examples happening around you?

Solutions

Sure, here's an explanation of each concept:

1. Guarantees or Warranties: Guarantees or warranties can be used to provide buyers with assurance that they are purchasing a high-quality product. For example, a car manufacturer may offer a warranty on a new car to guarantee its quality and reliability.
2. Liability Laws: Liability laws can help to hold sellers accountable for selling low-quality products. For example, if a seller knowingly sells a defective product that causes harm to the buyer, they may be held liable for damages.
3. Reputation: A seller's reputation can serve as a signal of the quality of their products. If a seller has a good reputation for providing high-quality products, buyers are more likely to trust them and make a purchase.
4. Experts: Experts can help to provide buyers with information about the quality of products. For example, a car mechanic can inspect a used car and provide a report on its condition, which can help to inform the buyer's decision.
5. Standards and Certification: Standards and certification can be used to ensure that products meet a certain level of quality. For example, a product may be certified by an independent organization to indicate that it meets certain quality standards.

Overall, these concepts can help to reduce the problem of asymmetric information in markets and increase the efficiency of resource allocation.

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Incomplete information about price

- Firms can obtain market power from consumers' lack of knowledge about prices. If one store raises its price above others and all consumers know it, consumers have **full information**, that store loses all its business. The market price is the competitive price p_c. In contrast, suppose that some or all customers do not know that other stores charge lower prices, consumers have **limited information** about price, a store can raise its price without losing all its sales. As shown in the following model, if there is a single price in the market, it is higher than p_c. However, there may be either no market or a multiple-price equilibrium (stores sell at different prices).



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The Tourist-Trap Model

- A typical tourist, Phoebe, arrives in a small town filled with souvenir stands. Each stand sells mugs with the town hall painted on it. Phoebe wanders by one of these sands, sees some mugs, and decides to buy one. She has but a short time before her bus leaves, and she does not expect to return to this town again. Thus, she does not have time to check the prices at each souvenir stand, and she cannot use information obtained through even a limited search in the future.

The Tourist-Trap Model

- If there are many such tourists, what prices do the stands charge for these mugs? To answer this question, we make four assumptions for specificity:
  - All firms (souvenir stands) have the same costs and sell the identical product.
  - All consumers have identical demand functions.
  - A guidebook provides each consumer with the general distribution of prices(how many stands charge each price), but does not give the particular price each stand charges.
  - The tourist's cost of going to a stand to check the price or to buy is c, which reflects the tourist's time and expenses (taxi rides).
- Thus, if Phoebe goes to two souvenir stands, her search costs are $2c$, If she buys a mug at the second stand at price p, her total cost is $p + 2c$. The least a mug will cost her is $p + c$, because she must visit at least one stand to buy a mug.



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The Tourist-Trap Model

- Due to symmetry, it is conceivable that all stands charge the same price. Do they have incentives to deviate when they all charge $p = p_c$?
- Suppose a stand charges $p = p_c + \epsilon$ with $\epsilon < c$. Would a tourist leave if she visits the stand?
- Now every stand has an incentive to raise its price to $p_c + \epsilon$. Do they still have incentives to deviate?
- The process continues, and every stand sets its price at the monopolistic price p_m. Does that constitute an equilibrium?



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The Tourist-Trap Model

- There are different possibilities. For instance, consider the possibility of a stand deviating and reducing its price by more than c.
- If there are only a small number of stands, the chance for a consumer to find the low-priced stand is large and they may be willing to do so. From the stands' perspective, they may aim for greater volume at the expense of per-sale profit. Hence, there is no single-price equilibrium.
- If there are a large number of stands, the chance for a consumer to find the low-priced stand is small and they may be unwilling to search for it. From the stands' perspective, they have less incentive to attract more consumers. Hence, a single-price (p_m) equilibrium emerges.



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Further extension

- Is it possible for someone to reduce the price from p_m by less than c?
- What determines the selection of equilibria other than the number of firms in the industry? The shape of consumer demand curves? The value of the search costs?
- What happens if the market allows free entry?
- Notice that, the main conclusions in the model remains irrespective of the value of c...
- Let us take a re-visit to how vulnerable the Bertrand Model is:
  - Capacity constraint
  - Product differentiation
  - Information

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The Tourist-Native Model

- Our analysis of the tourist-trap model raises two questions about markets in which consumers have limited information about prices.
- First, is there a model in which a multiple-price equilibrium is possible? That is, is there an equilibrium where stores charge different prices for the identical good, resulting in price dispersion?
- Second, if some consumers are fully informed while others have limited information, can there be a full-information equilibrium where the price equals the marginal cost?



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The Tourist-Native Model

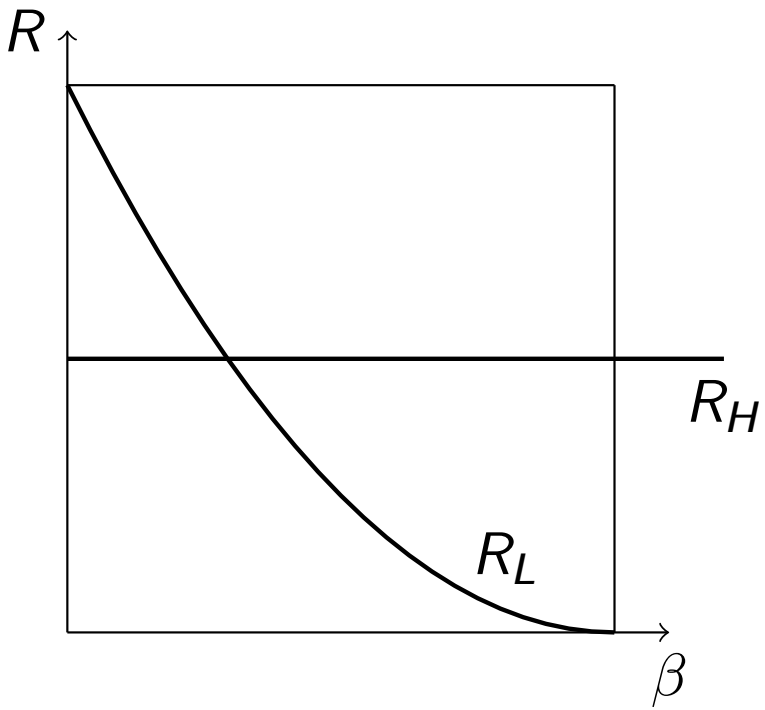
- In this model, there are two types of consumers with different search costs.
- Natives, who make up a proportion of α of the market, are informed consumers with zero search costs. They know the entire distribution of prices in the market, including which firm charges which price. Therefore, they will only visit the stand with the lower price p_l.
- Tourists, who make up a proportion of $1-\alpha$ of the market, are uninformed consumers with positive search costs. They will randomly visit the stand with the lower price p_l or the higher price p_h.
- Suppose that a proportion β of firms charge the low price p_l, firms can freely switch between low and high prices.



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Equilibrium

- For all low-priced stands: $TR_L = (\alpha L + (1-\alpha)L\beta)p_l$
- For all high-priced stands: $TR_H = (1-\alpha)L(1-\beta)p_h$
- A low-priced stand's revenue: $R_L = \frac{\alpha L+(1-\alpha)L\beta}{\beta n} p_l$
- A high-priced stand's revenue: $R_H = \frac{(1-\alpha)L(1-\beta)}{(1-\beta)n} p_h$
- When $R_L < R_H$, more stands would charge higher price, β decreases. Vice versa.



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Equilibrium

- In equilibrium, $\frac{\alpha L+(1-\alpha)L\beta}{\beta n}p_l = \frac{(1-\alpha)L(1-\beta)}{(1-\beta)n}p_h \rightarrow \beta^* = \frac{\alpha p_l}{(1-\alpha)(p_h-p_l)}$
- In the figure, we have $0 < \beta^* < 1$. Is it possible that we have $\beta^* \leq 0$ or $\beta^* \geq 1$? How to interpret them?

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The Tourist-Native Model

- Now, what happens if stands are cooperating instead of competing with each other?
- There are three possible strategies when all the stands are owned by a monopolist: charge unique low price ($R_l = Lp_l$), charge both prices (in which case it is optimal to only let one stand charge low price) ($R_{l\&h} = \alpha Lp_l + (1-\alpha)L(\frac{1}{n}p_l + \frac{n-1}{n}p_h)$), charge unique high price ($R_h = (1-\alpha)Lp_h$).
- Strategy one is clearly worse than strategy two, so it is a comparison between strategy two and three: a trade-off between losing the local consumer and charging higher prices for the tourists in the one stand.



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The Tourist-Native Model

$$\alpha L p_l + (1-\alpha) L(\frac{1}{n} p_l + \frac{n-1}{n} p_h) \longleftrightarrow (1-\alpha) L p_h$$

$$(\alpha + \frac{1-\alpha}{n}) p_l \longleftrightarrow (1-\alpha)(1 - \frac{n-1}{n}) p_h$$

$$(n\alpha + 1 - \alpha) p_l \longleftrightarrow (1-\alpha) p_h$$

$$\frac{n\alpha}{1-\alpha} + 1 \longleftrightarrow \frac{p_h}{p_l}$$

- When $\frac{p_h}{p_l}$ increases...
- When α increases...
- When n increases...
- Just for fun: what if the market allows free entry?

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Asymmetric information: A simple example

- Peter (Horizontal player 1) and Peppa (Vertical player 2) have a complicated relationship. If it is good, they want to see each other (coordination game). If it is bad, Peppa wants to avoid seeing Peter (matching pennies game).

	High	Low
High	2, 1	0, 0
Low	0, 0	1, 2

state G $(p = \frac{1}{3})$

	High	Low
High	2, 0	0, 2
Low	0, 1	1, 0

state B $(q = \frac{2}{3})$

- Peter does not know the relationship status, but Peppa does: and this is a common knowledge.
- We discuss three possible scenarios:
  - Scenario 1: they cannot observe each other's choice
  - Scenario 2: only 1 can observe 2's choice
  - Scenario 3: only 2 can observe 1's choice



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Scenario 1 Players' strategies

- Player 1 does not know the state and he does not know player 2's choice. All he can do is to choose between the two options: $\begin{cases} h \\ l \end{cases}$.
- Player 2 knows the state but she does not know player 1's strategy. So she can make her choice depending on the state, generating four possible strategies: $\begin{cases} G_h B_h \\ G_h B_l \\ G_l B_h \\ G_l B_l \end{cases}$.

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Scenario 1 Payoff table

	High	Low
High	2, 1	0, 0
Low	0, 0	1, 2

state G $(p = \frac{1}{3})$

	High	Low
High	2, 0	0, 2
Low	0, 1	1, 0

state B $(q = \frac{2}{3})$

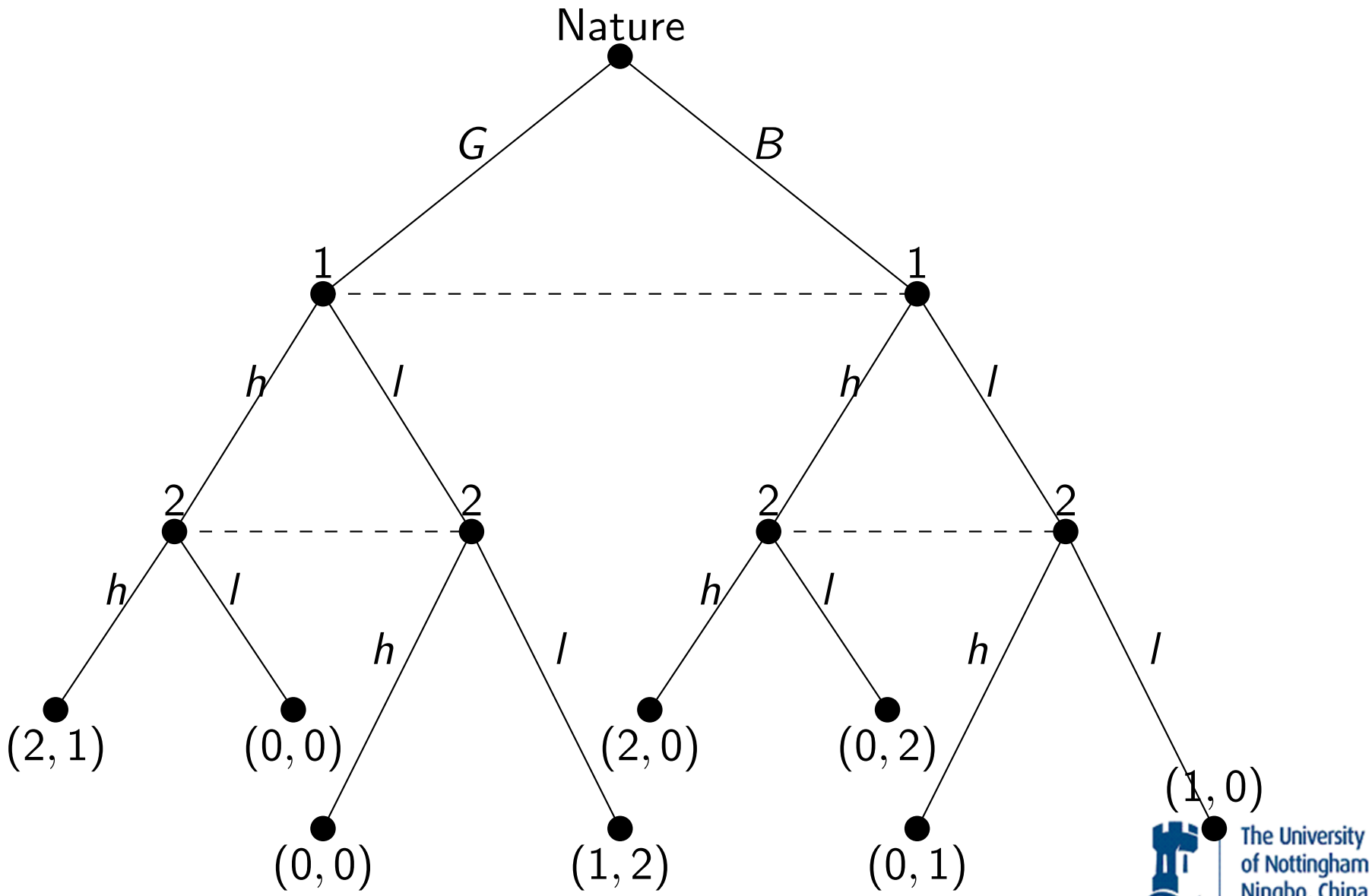
- Convert the two original payoff tables into a new one:

	GhBh	*GhBl*	*GlBh*	*GlBl*
h	$2p + 2q, 1p + 0q$	$2p + 0q, 1p + 2q$	$0p + 2q, 0p + 0q$	$0p + 0q, 0p + 2q$
l	$0p + 0q, 0p + 1q$	$0p + 1q, 0p + 0q$	$1p + 0q, 2p + 1q$	$1p + 1q, 2p + 0q$

	GhBh	*GhBl*	*GlBh*	*GlBl*
h	$2, \frac{1}{3}$	$\frac{2}{3}, \frac{5}{3}$	$\frac{4}{3}, 0$	$0, \frac{4}{3}$
l	$0, \frac{2}{3}$	$\frac{2}{3}, 0$	$\frac{2}{3}, \frac{4}{3}$	$1, \frac{2}{3}$

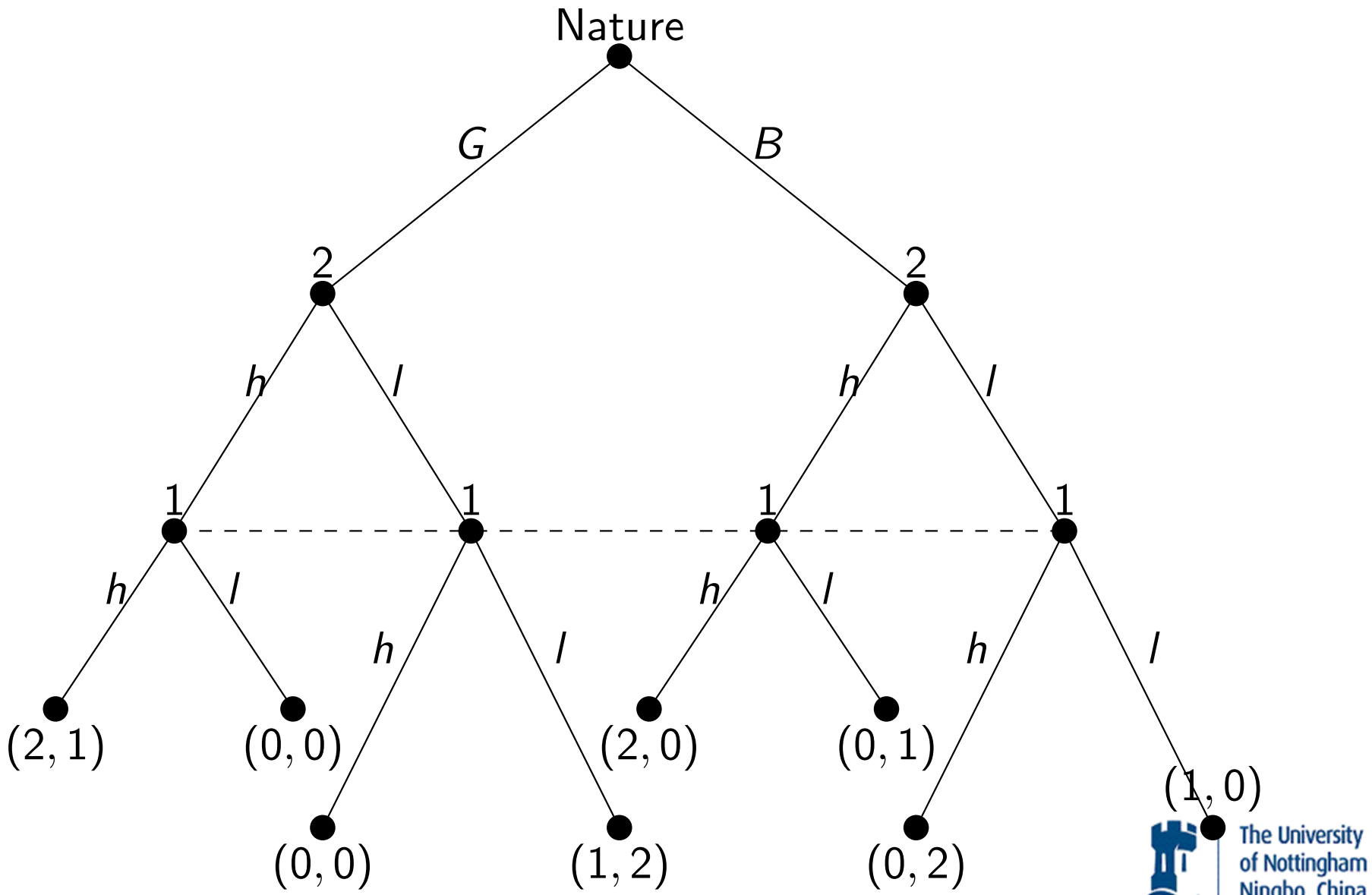
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Scenario 1 Game tree



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Scenario 1 Game tree



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Scenario 2 Players' strategies

- Player 1 does not know the state but he knows player 2's choice. So he makes his choice dependent on player 2's,

  generating four possible strategies: $\begin{cases} h_h l_h \\ h_h l_l \\ h_l l_h \\ h_l l_l \end{cases}$.

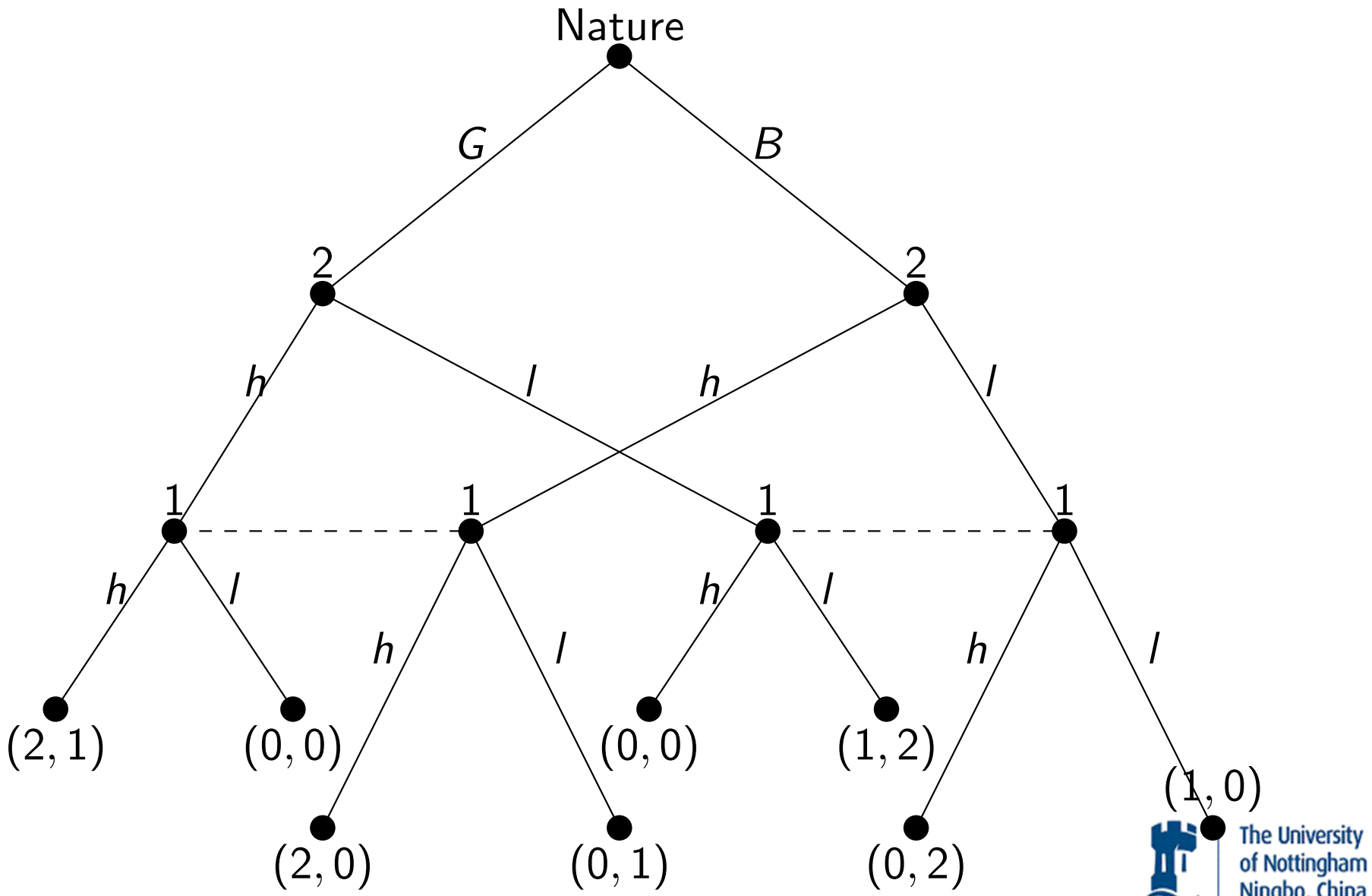
- Player 2 knows the state but she does not know player 1's strategy. So she can make her choice depending on the state,

  generating four possible strategies: $\begin{cases} G_h B_h \ (pooling) \\ G_h B_l \ (separating) \\ G_l B_h \ (separating) \\ G_l B_l \ (pooling) \end{cases}$.



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Scenario 2 Game tree



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Scenario 2 Signals

- Scenario 2 is a typical "signaling game". Where is the "signal"?
- Suppose that 2 chooses G_hB_h, what is the signal 1 receives seeing h?
- Suppose that 2 chooses G_hB_l, what is the signal 1 receives seeing h?
- Suppose that 2 chooses a more complicated strategy: when the state is G, 2 chooses h with probability m and when state is B, 2 chooses h with probability n. What is the signal 1 receives seeing h now?



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Scenario 2 Bayes' rule

- The probability that the state is G and 1 receives h: $\frac{1}{3}m$.
- The probability that the state is B and 1 receives h: $\frac{2}{3}n$.
- The probability that 1 receives h: $\frac{1}{3}m + \frac{2}{3}n$.
- The probability that the state is G when 1 receives h: $\frac{\frac{1}{3}m}{\frac{1}{3}m+\frac{2}{3}n}$.

Similarly:

- The probability of B when 1 receives h is $\frac{\frac{2}{3}n}{\frac{1}{3}m+\frac{2}{3}n}$.
- The probability of G when 1 receives l is $\frac{\frac{1}{3}(1-m)}{\frac{1}{3}(1-m)+\frac{2}{3}(1-n)}$.
- The probability of B when 1 receives l is $\frac{\frac{2}{3}(1-n)}{\frac{1}{3}(1-m)+\frac{2}{3}(1-n)}$.



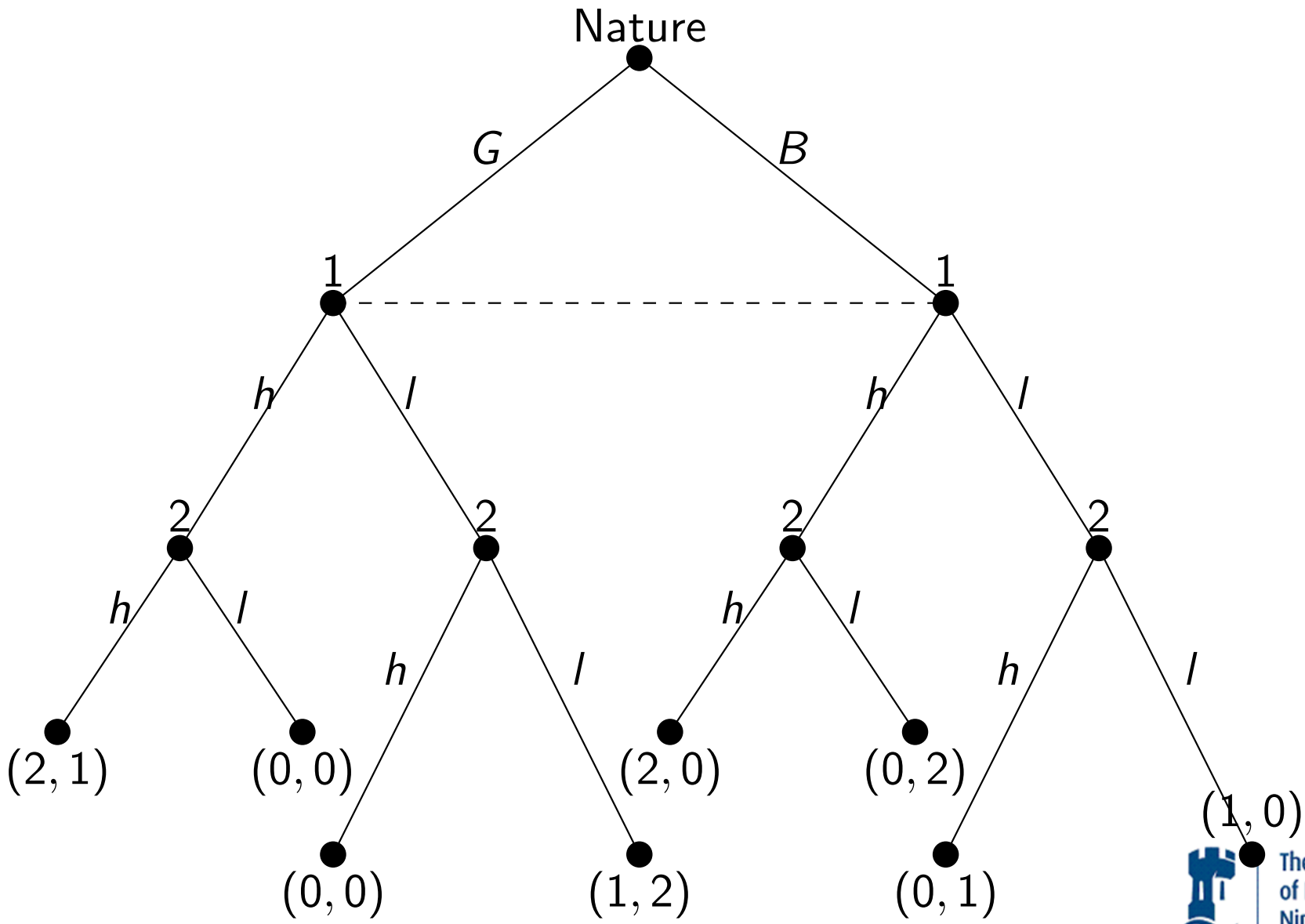
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Scenario 3 Players' strategies

- Player 1 does not know the state or player 2's choice. All he can do is to choose between the two options: $\begin{cases} h \\ l \end{cases}$.
- Player 2 knows the state and player 1's strategy. So she can make her choice accordingly.

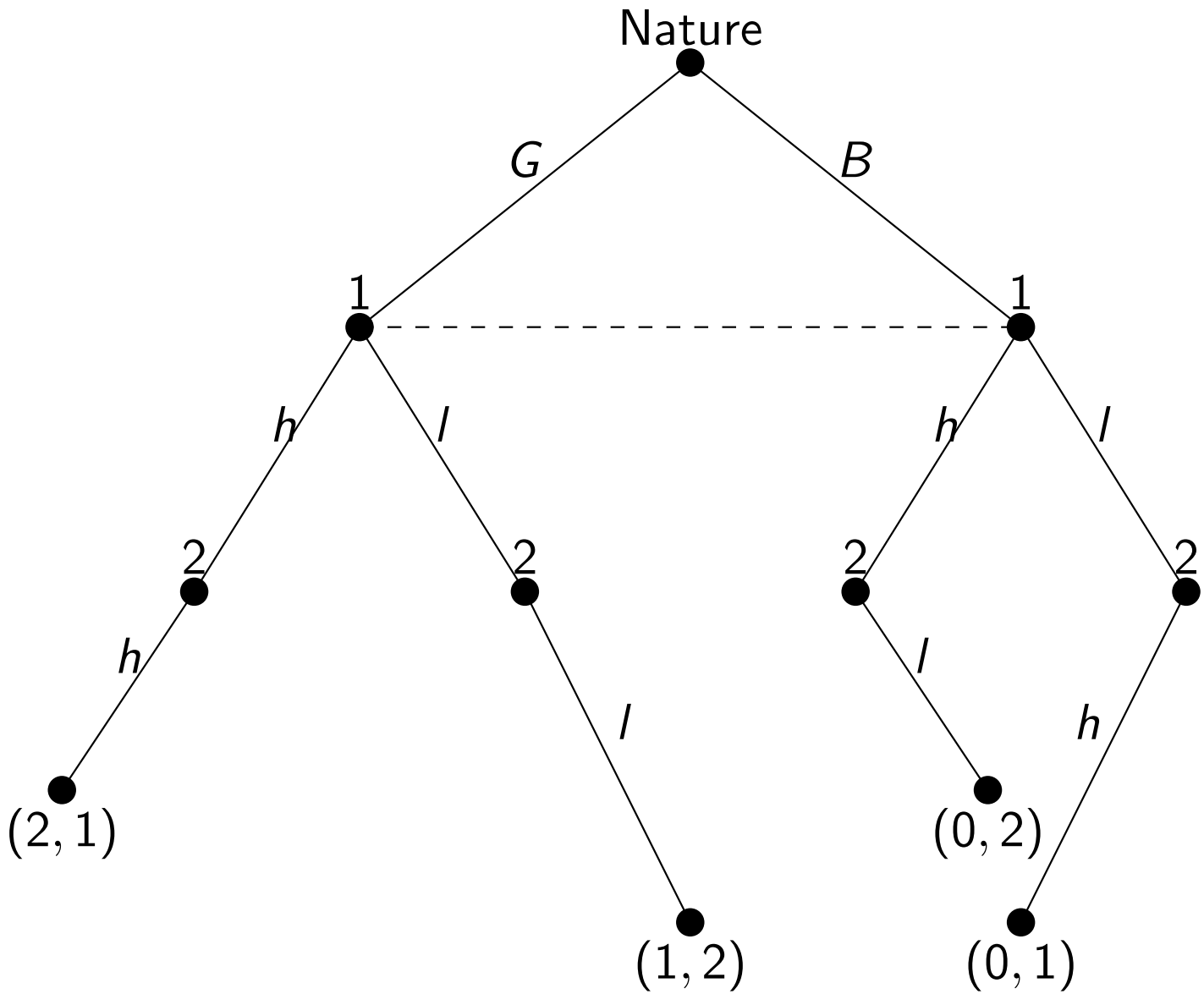
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Scenario 3 Game tree



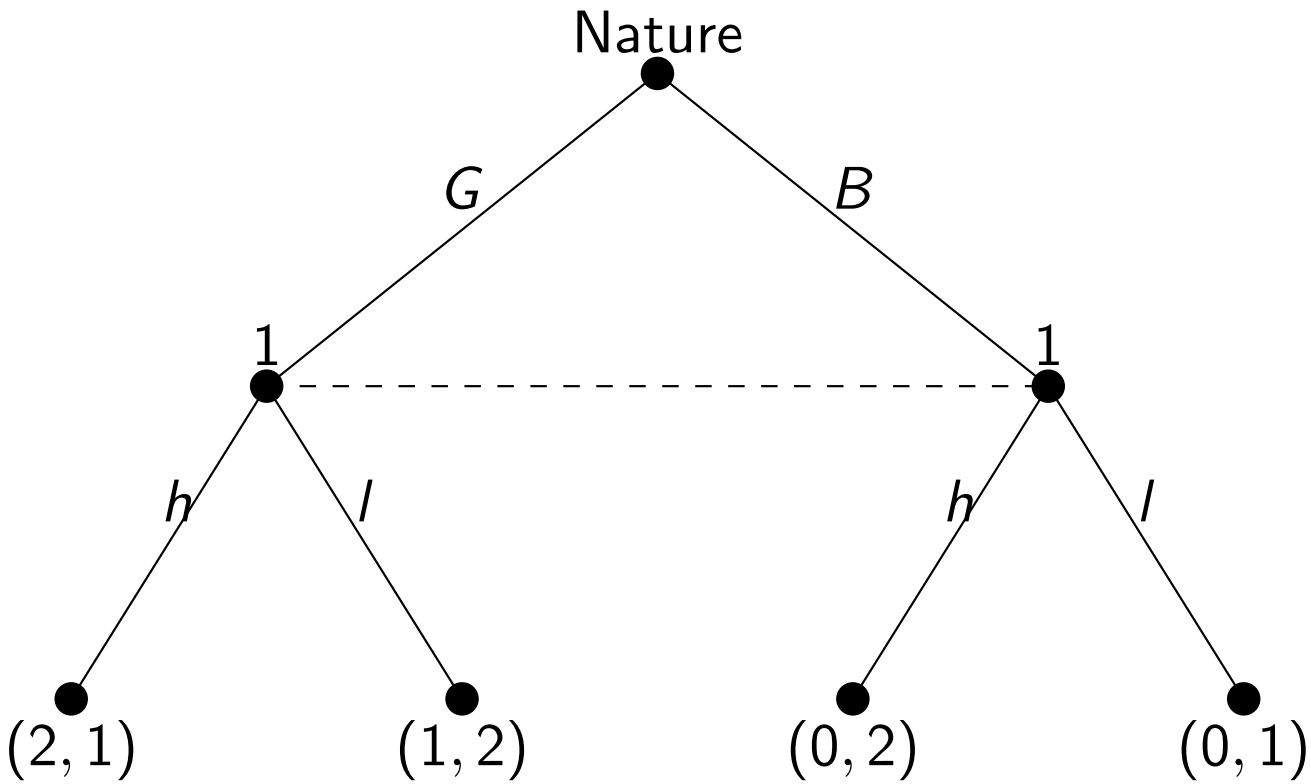
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Scenario 3 Sub-game perfect equilibrium



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Scenario 3 Sub-game perfect equilibrium



- 1's (expected) payoff from h: $2 * \frac{1}{3} + 0 * \frac{2}{3}$
- 1's (expected) payoff from l: $1 * \frac{1}{3} + 0 * \frac{2}{3}$

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Two Primary Ways to Influence Agents' Behaviors

- Design/provide incentives (Mechanism Design)
  - Auctions
  - Discounts/coupons
  - Job contract design
- Influence agents' beliefs
  - Lie/Cheap Talk (unverifiable information)
  - Persuasion games (verifiable information)
  - Bayesian persuasion



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Example: Recommendation Letters

- Background story: all UNNC students ranked purely according to their GPA, from high to low. The top 30% of UNNC students (say, those with a distinction degree) can survive in Oxford.
- The admissions officer from Oxford, let us call him James: if he approves a qualified (the top 30%) student or rejects an unqualified one, his KPI increases by 1, Otherwise, decreases by 1.
- When I am writing a recommendation letter for UNNC students, my KPI is to maximize their chance of being accepted: no matter qualified or not.
- What should I do?

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Strategy 1: No information

- Suppose that I do nothing (provide a "template" recommendation letter for every one). Finally, my RL will be ignored.
- For James, he makes his decision based on his expected payoff: $U_{approve} = 1 * 0.3 - 1 * 0.7 = -0.4$ and $U_{reject} = -1 * 0.3 + 1 * 0.7 = 0.4$. He rejects every application from UNNC.

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Strategy 2: False information

- Suppose that I deliberately provide the false information, give a "distinction" to everybody?
- In the long term James would realize the problem and my recommendation would be no longer considered, and I end up helping no one.
- In this case, false information = no information.

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Strategy 3: True information

- I can be completely truthful and build up my reputation in the long run. In this case, I get 30% of UNNC students admitted into Oxford.

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Strategy 4: Persuasive information

- Suppose that I "decorate" the information a bit, say, if you are at 50%, I say you are the toppest 50%?
- Actually that is what we do all the time. But James is smart enough to combine one with one, so it is equivalent to Strategy 3.

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Strategy 5: Yes and No



The art of language: "Yes Minister"

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Strategy 5: Yes and No

- Suppose that I do this: I "blur" my recommendations.
- For the 30% qualified ones $(+)$, I give a high recommendation.
- For the 70% unqualified ones $(-)$, I give a low recommendation.
- Does James like it now?
- What if I start to "squeeze" one unqualified into the qualified?
- What about another one?

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Strategy 5: Yes and No

- Ultimately:
- For the 30% qualified ones $(+)$, I give a high recommendation.
- For the 70% unqualified ones $(-)$, I give a high recommendation with a chance of $p(h|-) = \frac{3}{7}$, and a low recommendation with a chance of $p(l|-) = \frac{4}{7}$.
- Basically, I claim that I believe all first degree and second degree should get into Oxford, I refuse to divide them and they all have my highest recommendation.
- Why do I do so?

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Strategy 5: Yes and No

- Even if James perfectly understands my incentive and my strategy (I would actually prefer he understands), so that he knows I am trying to "oversell" UNNC students, let's see what he is dealing with:
- When he receives a low recommendation, he knows it comes from an unqualified student and he rejects the application.
- When he receives a high recommendation, he knows it might come from the unqualified student, or come from the qualified student. What is the chance of each?



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Strategy 5: Yes and No

- The chance of James receives a h from an unqualified student: $p(-) * p(h|-) = 0.7 * \frac{3}{7} = 0.3$
- The chance of James receives a h from a qualified student: $p(+) * p(h|+) = 0.3 * 1 = 0.3$.
- According to Bayesian Persuasion, given that he receives a h, the chance that it comes from a qualified student: $p(+|h) = \frac{p(+)*p(h|+)}{p(+)*p(h|+)+p(-)*p(h|-)} = 0.5$.
- EP(approving)$=(p(+|h) * 1 - p(-|h) * 1)p(h) + p(l) * 1 = 0.4$
- EP(rejecting)$=(-p(+|h) * 1 + p(-|h) * 1)p(h) + p(l) * 1 = 0.4$
- Très bien. He has no choice but to approve the application!

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Strategy 5: Yes and No

- So what is the outcome of my beautiful strategy?
- James rejects all applications with l recommendation but approves all application with h recommendation.
- Notice that I give l with a percentage of $p(l) = p(-)p(l|-) = 0.7 * \frac{4}{7} = 0.4$, and I give h recommendation with a percentage of $p(h) = p(-)p(h|-) + p(+)p(h|+) = 0.7 * \frac{3}{7} + 0.3 * 1 = 0.6$.
- Thanks to my contribution, I get 60% of UNNC students admitted into Oxford!

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The interpretation

- Am I lying? Or am I being truthful?
- Even if you consider concealing part of the truth to be equal to lying, what if I really cannot tell the students above upper second?
- Should we encourage everyone to try Oxford?
- Intuitively, how and why does Bayesian persuasion work?
- Some other examples?



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Thank you





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